toward one another. Crucial to this effect, though, is that the baseballs were released in slightly different locations in space, so that their freely falling paths toward earth's center were slightly different as well. Thus, a more precise statement of Einstein's realization is that the smaller the spatial extent of an object, the more fully it can eliminate gravity by going into free fall. While an important point of principle, this complication can be safely ignored throughout the discussion.

15. For a more detailed, yet general-level, explanation of the warping of space and time according to general relativity, see, for example, Chapter 3 of *The Elegant Universe*.

16. For the mathematically trained reader, Einstein's equations are $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$, where the left-hand side describes the curvature of spacetime using the Einstein tensor and the right-hand side describes the distribution of matter and energy in the universe using the energy-momentum tensor.

17. Charles Misner, Kip Thorne, and John Archibald Wheeler, *Gravitation* (San Francisco: W. H. Freeman and Co., 1973), pp. 544–45.

18. In 1954, Einstein wrote to a colleague: "As a matter of fact, one should no longer speak of Mach's principle at all" (as quoted in Abraham Pais, *Subtle Is the Lord,* p. 288).

19. As mentioned earlier, successive generations have attributed the following ideas to Mach even though his own writings do not phrase things explicitly in this manner.

20. One qualification here is that objects which are so distant that there hasn't been enough time since the beginning of the universe for their light—or gravitational influence—to yet reach us have no impact on the gravity we feel.

21. The expert reader will recognize that this statement is, technically speaking, too strong, as there are nontrivial (that is, non–Minkowski space) empty space solutions to general relativity. Here I am simply using the fact that special relativity can be thought of as a special case of general relativity in which gravity is ignored.